Clean Accuracy Is Not a Diagnosis: A Noise-Stress Re-evaluation of Lightweight Bearing Fault Diagnosis

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Abstract

Lightweight one-dimensional convolutional networks are the default choice for bearing fault diagnosis on small hardware, and most are tested on the public CWRU (Case Western Reserve University) benchmark. Clean accuracy on CWRU has been close to 100% for years, so the benchmark has almost no resolution left: a saturated metric cannot tell a robust model from a fragile one. We re-evaluate eight lightweight architectures (WDCNN, TICNN, MCNN, ResNet1D, DS-CNN, LeNet1D, ViT1D and MLP-Mixer1D) under a leakage-safe protocol with no shared sampling points between the training and test windows. Clean-test accuracy stays within three points (97.5–100%), and overlapping-window leakage raises it by at most 0.63 points. The same models differ by up to 55 points when the test signal carries additive noise at 10 dB SNR, and the clean ranking does not predict the noisy one. Much of this brittleness comes from the training recipe: drawing a random SNR for each mini-batch lifts mean 10 dB accuracy from 0.70 to 0.97 and the weakest model from 0.43 to 0.95, at a cost of 5.4 clean points. A multi-channel wide first-layer stem adds 13–52 points at 10 dB for 0.5–3% more parameters, and the number of channels matters more than the kernel width. Open-set rejection depends on the architecture and is moderately anti-correlated with noise robustness ($\rho = -0.64$, $p = 0.09$); a Mahalanobis distance score recovers most of the rejection that augmentation costs (mean area under the receiver-operating-characteristic curve (AUROC) 0.80 to 0.94). The collapse and its repair reproduce on Paderborn, but the ranking changes under impulsive, harmonic and drift noise, so a robustness claim that does not name the perturbation cannot be compared. We release the protocol, the split files and every per-seed result.

Keywords: bearing fault diagnosis; lightweight neural networks; noise robustness; leakage-safe evaluation; open-set recognition; reproducibility

1 Introduction

Rolling-element bearing faults are one of the most common causes of failure in rotating machinery, and vibration-based diagnosis is the usual non-invasive way to monitor them [1]. Deep learning is now the standard tool for the job: a one-dimensional convolutional network (1D-CNN) takes a short window of the accelerometer signal and outputs a fault class. Because the models and the datasets are both small, papers in this area tend to follow a similar pattern: a new block, attention module or multi-scale fusion path is proposed, trained on CWRU, and reported with a table of accuracies that now often reach 99% or even 100% [2–5].

This way of reporting has a simple problem: a benchmark on which every method scores 99% cannot be used to choose a method. If WDCNN reaches 99.7% and our ResNet1D reaches 99.98%, the difference is 0.3 points, far below the effect of changing the split protocol, yet it is often reported as an improvement. The numbers are also hard to compare across papers, because split protocols differ: some authors split

*Code, split files and per-seed results are available at https://github.com/jinchi2007/clean-acc-not-diagnosis.

overlapping windows at random, some split by recording, and some do not say [6, 7]. Recent position papers have pointed out these evaluation problems, but as far as we know the size of the effect has not been measured across a common set of architectures under one harness.

The same literature makes a second, largely separate claim: that the proposed models are "robust to noise". Robustness is usually shown by adding synthetic noise at a few signal-to-noise ratios (SNRs), white Gaussian in most papers and coloured in others, and by showing that the new model degrades more gracefully than its predecessors [8–10]. Because the SNRs, the noise model and the training procedure change from paper to paper, these claims are equally hard to compare.

We therefore built one evaluation harness. Eight lightweight architectures—chosen to cover the design space of the field, from a 3k-parameter LeNet to a 750k-parameter MLP-Mixer—are trained with the same hyperparameters and evaluated on the same splits under (i) four split protocols, (ii) five test SNRs plus the clean signal, (iii) with and without noise augmentation during training, and (iv) an open-set setting in which three fault classes are withheld from training. A second dataset (Paderborn) checks that the conclusions are not just an artefact of CWRU.

Leakage turns out to be small: at full data the leaky and the leakage-safe splits differ by at most 0.63 points. Noise is not small. At 10 dB the same eight models span 55 points, a range that the clean benchmark hides, and the two rankings are weakly anti-correlated (Spearman $\rho = -0.38$). A one-line change to the training loop, drawing a random SNR for each mini-batch, recovers most of the lost accuracy for every architecture we tried, at a cost of about 5 clean points.

This paper makes five contributions:

1. **A saturation audit.** Under a leakage-safe protocol (per-recording chronological split with a $w/2$ guard band, zero shared sampling points, checked by hashing), the eight lightweight models span only 2.5 points of clean CWRU accuracy. The inflation from overlapping-window leakage is at most 0.63 points at full data and 0.92 points when only 20 windows per class are used, while changing the split protocol moves accuracy by an order of magnitude more.

2. **A noise-stress matrix.** The same models, trained the same way, differ by up to 55 points at 10 dB SNR, but by only 3 points on clean data in the three-seed matrix; the clean ranking does not predict the noisy ranking (Spearman $\rho = -0.38$, $p = 0.35$, $n = 8$).

3. **An attribution study.** By adding a wide-kernel front-end to narrow-kernel backbones and by adding per-batch noise augmentation, we separate architectural effects from training-recipe effects. Each removes much of the gap, through a different mechanism, and the two are complementary.

4. **An open-set audit.** With three of ten classes withheld, closed-set accuracy stays near 100% while rejection quality varies from 0.12 to 0.87 true-positive rate (TPR) at a 5% false-positive rate (FPR) across architectures, and it is anti-correlated with noise robustness ($\rho = -0.64$)—a deployment risk that closed-set tables cannot show. Under the softmax read-out, the augmentation that repairs noise accuracy does not repair rejection: mean AUROC falls from 0.90 to 0.80. Most of the loss is in the read-out: with a Mahalanobis distance score instead of the softmax maximum, the same networks reach 0.95 (clean) and 0.94 (augmented), so the two axes have to be reported separately, and the rejection score named together with the open-set number.

5. **A reproducible protocol and reporting checklist** (Appendix A), including the exact split files, the per-seed results and a seven-item list of information we think every diagnostic paper should report.

This paper proposes no new architecture and uses only public datasets. Its claim is about evaluation: *clean accuracy is not a proxy for diagnostic capability, and the usual protocol cannot see the difference.*

2 Related work

Lightweight 1D architectures. WDCNN [2] established the wide-first-kernel design that most subsequent work follows: a first layer with a kernel of 64 samples and a stride of 16 acts as a learnable band-pass filter and reduces temporal resolution before the deep stack. TICNN [8] adds ensemble training and

dropout; MCNN [3] uses multi-scale kernels; ResNet1D, DS-CNN and LeNet1D are imported from the general 1D-CNN and keyword-spotting literature [11, 12]. More recent work replaces convolution with attention or mixing blocks, e.g. ViT-style [13] and MLP-Mixer-style [14] backbones, or with graph, contrastive, domain-adversarial or imputation machinery for cross-domain diagnosis [15–17]. We include one representative of each family so that conclusions are not tied to a single design lineage.

Noise robustness. Adding synthetic noise at test time is the usual robustness probe, but the protocol differs from paper to paper: some add white Gaussian noise at a handful of SNRs [8, 18], others train against coloured noise or in noisy working conditions [9, 10], and the SNR grids rarely agree. Reported robustness therefore cannot be compared across papers, and a model robust to white noise may fail on the impulsive noise of a real gearbox. The main matrix uses white Gaussian noise; Section 4.9 repeats a subset with three other families.

Evaluation methodology. Benchmark practice has been critiqued in adjacent fields, but for vibration diagnosis specifically, recent position papers flag overlapping-window leakage and non-comparable splits [6, 7], and recent reviews document the growth of the field and its open challenges [19]. We are not aware of a study that quantifies the effect across a common architecture set under a single harness; our audit is designed to be reusable, since it needs only hashes of the window index sets.

Open-set diagnosis. Most papers assume the test set contains only fault classes seen in training. In practice a machine can develop a fault that was never labelled, and a confident wrong answer is worse than an abstention. Open-set recognition is well studied in vision but rare in vibration diagnosis, and where it appears it is evaluated with different metrics. We report AUROC and TPR at a fixed 5% false-acceptance rate.

3 Protocol

3.1 Data

CWRU [20]: 40 recordings (10 classes: normal plus three fault diameters $\times$ three fault locations, 0.007–0.021 in), four load conditions (0–3 hp), 12 kHz drive-end accelerometer, 1024-sample windows with 50% overlap. Ten classes are used throughout.

Paderborn [21] is used as a second dataset: six healthy and twelve artificially damaged bearing settings at 64 kHz, 1,439 records of 4 s each, aggregated into two classes (healthy, outer race) with the same windowing, split and training logic. The twelve damaged bearings in the released subset all carry outer-race damage; inner-race settings exist in the database but are not part of our subset, so the Paderborn experiment tests generalisation of the *evaluation method* rather than a three-class diagnosis problem.

3.2 Splits

Four protocols, all released as index files:

- **leaky**: pool all windows, stratified random 80/20 split by class. Overlapping windows share samples across the split.
- **safe**: for each recording the first 70% of windows go to training and the last 30% to test, with a half-window guard band; no sampling point appears in both sets (verified by hashing the sample-index sets).
- **crossload**: train on 0 and 1 hp, test on 2 and 3 hp.
- **filedisjoint**: 25% of the recordings per class are held out entirely.

3.3 Models and training

Eight architectures, 3,018–747,722 parameters, trained from scratch with AdamW (lr $= 2 \times 10^{-3}$, weight decay 10^{-4}), cosine schedule, batch 128, 20 epochs (15 for Paderborn), one to three seeds depending on the experiment (three for the CWRU noise matrices, the four-quadrant comparison and the open-set audit of Section 4.6; two for the remaining audits; the front-end ablation of Table 5 is a single seed at 15 epochs), per-window z-score normalisation, no pre-training. All models use the same optimiser budget, so the differences are not explained by unequal tuning effort. Training runs on Apple MPS; a CWRU run takes seconds to a few minutes, a Paderborn run 2–8 minutes.

3.4 Noise and augmentation

Test noise is additive white Gaussian at SNR $\in \{15, 10, 5, 0, -5\}$ dB, applied after per-window z-score normalisation; "clean" corresponds to the uncorrupted window. Clean-trained models see no noise during training; augmented models draw SNR $\sim U[0, 15]$ dB independently per mini-batch.

To test whether the conclusions depend on the noise family, we additionally corrupt the test set with three additional families at the same nominal SNR: (i) *harmonic* interference, a sinusoid at a per-window frequency $f \in [0.02, 0.40]$ cycles/sample plus its $2.3f$ harmonic; (ii) *impulsive* noise, 16 impulses of amplitude 3 placed at uniformly random sample positions (one per 64 samples); (iii) *drift*, a single low-frequency sinusoid with $f \in [0.002, 0.02]$ cycles/sample and random phase. Each realisation is rescaled to the requested SNR with respect to the signal power of that window, so the four families are compared at equal SNR.

3.5 Open-set protocol

Three classes are removed from training, in two configurations: the three 0.021-in faults (classes 3, 6, 9) and the three 0.007-in faults (classes 1, 4, 7). The model is trained on the remaining seven and evaluated on a test set containing all ten. The rejection score is the softmax maximum in Sections 4.6–4.7, and Section 4.8 compares it with post hoc alternatives on the same networks (negative free energy and a Mahalanobis distance, the latter measured both in logit space and in penultimate-feature space). We report AUROC, TPR at 5% FPR on unknowns, and closed-set accuracy on the known classes.

3.6 Statistics

Means and standard deviations over two or three seeds (three for the CWRU noise matrices, the four-quadrant comparison and the open-set audit of Section 4.6; two for the remaining audits, including the decision-layer audits of Sections 4.7–4.8). Paired comparisons use the Wilcoxon signed-rank test over the $8 \times 3 = 24$ (model, seed) pairs for clean-versus-augmented training and 32 (model, holdout, seed) pairs for the open-set comparisons of Sections 4.7–4.8, with Holm correction. Rank agreement between clean and noisy rankings uses Spearman's ρ. Every table in this paper can be regenerated from `results/runs/` by the commands in Appendix A.

4 Results

4.1 Clean accuracy is saturated

Table 1 and Figure 1 show the four split protocols for all eight models. Under the leakage-safe protocol seven of the eight models exceed 99.6%, and the whole span is 2.5 points, from LeNet1D at 0.975 to MCNN at 1.000. In Figure 1 the two left columns look the same.

The split matters much more. The cross-load protocol costs 5 to 13 points per model (WDCNN 0.948, LeNet1D 0.856), about 10 points on average, and it is the only protocol that separates the models. WDCNN is the best there even though it is only fifth under the leakage-safe protocol. The three narrow-kernel

Table 1: Clean test accuracy (mean±std over two seeds, 20 epochs, 10-class CWRU) under four split protocols. *leaky*: random window-level split with 50% overlap between neighbouring windows; *safe*: per-recording chronological 70/30 split with a $w/2$ guard band and zero shared sampling points; *cross-load*: train on loads 0 and 1, test on loads 2 and 3; *recording-disjoint*: train and test on disjoint recordings. Best value per column in bold; columns whose maximum is shared by several models are left unbolded.

model	leaky	safe	cross-load	recording-disjoint
WDCNN	.999±.000	.997±.001	**.948**±.003	.955±.052
TICNN	1.000±.000	1.000±.000	.911±.013	.997±.005
MCNN	1.000±.000	1.000±.000	.900±.000	.997±.004
ResNet1D	1.000±.000	1.000±.000	.899±.000	.997±.002
DS-CNN	1.000±.000	1.000±.000	.900±.001	**.998**±.003
LeNet1D	.982±.003	.975±.002	.856±.002	.954±.048
ViT1D	.998±.001	.996±.002	.894±.009	.986±.012
MLP-Mixer1D	.999±.001	.996±.001	.870±.008	.953±.050
spread	1.8	2.5	9.1	4.5

convolutional backbones (MCNN, ResNet1D, DS-CNN) all land within half a point of 0.899, while WDCNN (0.948) and TICNN (0.911) sit clearly above it, so the convergence looks like a limitation of the narrow-kernel input layer rather than of the individual models. The recording-disjoint column lies in between and has the largest seed spread (WDCNN 0.955 ± 0.052), so we use it as a consistency check and not for ranking.

Leakage is a small effect. The leaky split is above the safe one for seven of the eight models (MCNN ties at 1.000), but by at most 0.63 points (LeNet1D 0.982 vs. 0.975) and by 0.3 points or less for six of them. Going from leaky to cross-load costs 5–13 points, an order of magnitude more than leakage costs at full data. We repeated the comparison in a 20-windows-per-class regime, where the overlap removes a much larger fraction of the training windows. There the mean leaky-minus-safe gap is $+0.44$ points (four models, two seeds, at most $+0.92$ for WDCNN). That regime collapses LeNet1D to 0.38 while MCNN stays at 0.98, so it exposes model differences that full data hides; leakage is still negligible.

Finding 1. All eight models reach at least 97.5% under a leakage-safe protocol and span 2.5 points, which is less than the effect of changing the split. Clean CWRU accuracy therefore cannot rank these architectures, and it cannot support the architectural claims made in the literature.

Finding 2. Overlapping-window leakage inflates accuracy by at most 0.63 points at full data and by at most 0.92 points with 20 windows per class. What separates published numbers is the split choice, not leakage; the bigger problem is saturation.

4.2 Noise stress: a 55-point spread and a rank inversion

Table 2 and Figure 2(a) evaluate the same eight models on noisy test signals, and Figure 3 (left) plots clean against 10 dB accuracy. The clean column spans 0.970 to 1.000, 3 points; the 10 dB column spans 0.427 to 0.980, 55 points.

Finding 3. At 10 dB SNR the eight architectures are separated by 55 points, while on clean data they are separated by 3. Seven of them reach at least 99.5% clean accuracy, and three of those fall below 60% at 10 dB. Across the eight models the Spearman correlation between clean and 10 dB accuracy is $\rho = -0.38$ ($p = 0.35$, $n = 8$), so the clean ranking has no reliable association with the noisy one and, if anything, points the wrong way. Only the two models with wide first-layer kernels (WDCNN, TICNN) survive.

4.3 Noise augmentation repairs most of the gap

Table 3 repeats the matrix with one change: during training each mini-batch draws a random SNR from $U[0, 15]$ dB. Figure 2(b) shows the resulting curves and Figure 3 (right) the per-model gain.

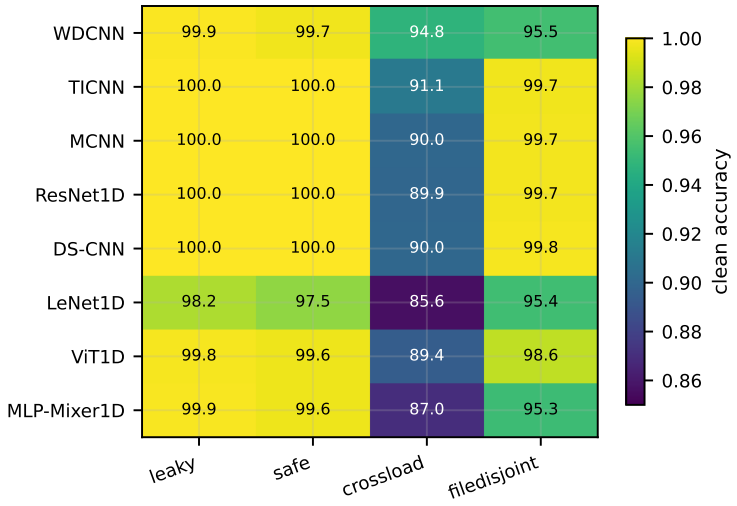


Figure 1: Clean test accuracy of eight lightweight models under four split protocols. Overlapping-window leakage (leftmost column) and the leakage-safe protocol are indistinguishable; the cross-load column is the only one that separates models.

Table 2: Clean-trained models, test accuracy (mean over 3 seeds) at five SNRs (plus clean). "Clean" is the uncorrupted test set. Standard deviations at 10 dB are given in parentheses; the highest and the lowest 10 dB entry are bold. The full per-seed table accompanies the released results.

model	params	clean	15 dB	10 dB	5 dB	0 dB	−5 dB
WDCNN	34,410	.9970	.9953	**.9803** (.010)	.8362	.4053	.1310
TICNN	83,914	.9998	.9998	.9571 (.019)	.6465	.2887	.1377
ViT1D	139,850	.9964	.9741	.7553 (.005)	.4675	.2705	.1294
MLP-Mixer1D	747,722	.9968	.9772	.7504 (.019)	.4433	.3051	.1999
LeNet1D	3,018	.9698	.9470	.7062 (.075)	.4080	.1690	.1067
MCNN	184,554	1.0000	.9315	.5856 (.038)	.2507	.1004	.1005
DS-CNN	69,642	.9995	.7887	.4281 (.016)	.2969	.2281	.1296
ResNet1D	241,322	.9999	.8051	**.4271** (.012)	.2646	.1357	.1294

Finding 4. Augmentation compresses the 10 dB spread from 55 to 13 points. The worst clean-trained model at 10 dB (ResNet1D, 0.427) rises to 0.945, the second worst (DS-CNN) to 0.969, and even the weakest augmented model reaches 87%. Averaged over the 24 (model, seed) pairs, 10 dB accuracy rises from .699 to .967 (+26.8 points, Holm-corrected $p < 10^{-6}$), 5 dB from .452 to .969 and 0 dB from .238 to .687 (Table 4). The price is 5.4 clean points on average: the four backbones that already mix information across the whole window (WDCNN, TICNN, ViT1D, MLP-Mixer1D) lose at most 1.3 points, while the narrow-kernel and multi-scale CNNs lose 8.8–12.7.

4.4 A multi-channel front-end at almost no clean cost

The previous two subsections suggest that robustness depends on what happens at the input: noise augmentation changes the input distribution during training, and wide first-layer kernels act as a fixed band-pass filter. To test the second mechanism directly we insert a learnable front-end in front of narrow-kernel backbones—c parallel convolutions with kernel k and stride $s = 16$ throughout, concatenated in the channel dimension—and sweep $(k, c) \in \{(32, 8), (64, 1), (64, 4), (64, 8), (128, 8)\}$ (Table 5, Figure 4).

Finding 5. The number of front-end channels is the dominant factor, although kernel width also matters: at $c = 8$, widening k from 32 to 64 adds 7.0 points at 10 dB and from 64 to 128 a further 2.7, while raising c from 1 to 8 adds 15.4. For ResNet1D, one wide channel ($k = 64, c = 1$) adds +34.3 points at

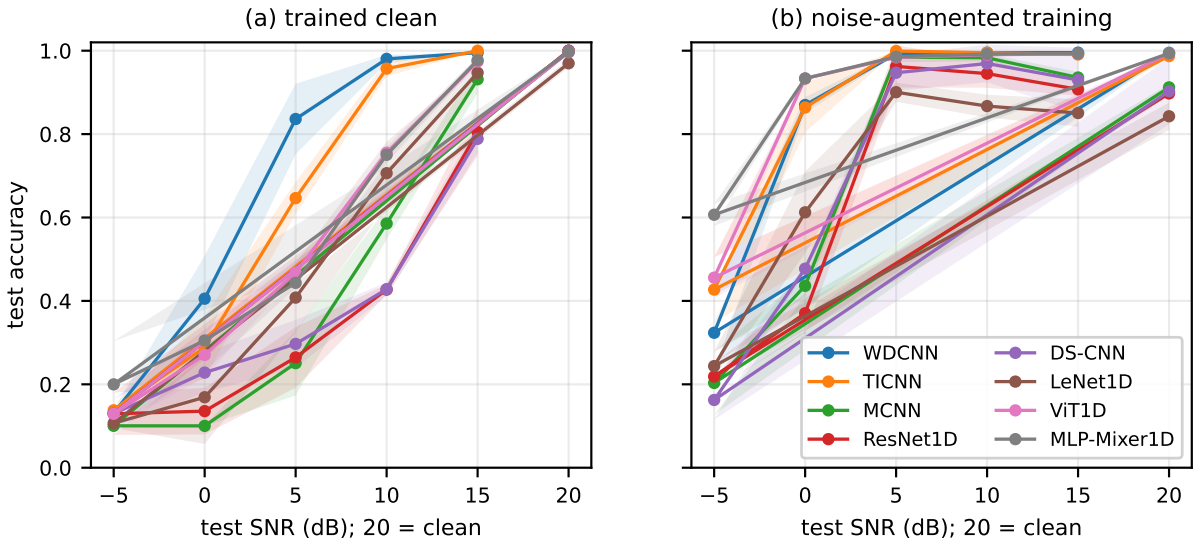


Figure 2: Test accuracy versus SNR for the eight architectures. (a) Trained on clean data: at 10 dB the models span 55 points and the ordering is unrelated to clean accuracy. (b) Trained with per-batch noise augmentation: at 10 dB the curves collapse into a band of 0.87–1.00, i.e. from a 55-point spread to 13 points. Shaded regions are ± 1 std over three seeds.

Table 3: As Table 2, but with noise augmentation (SNR $\sim U[0, 15]$ dB per mini-batch) during training.

model	clean	15 dB	10 dB	5 dB	0 dB	−5 dB
WDCNN	.9949	.9953	**.9953**	.9927	.8697	.3236
TICNN	.9872	.9909	.9951	.9991	.8638	.4271
ViT1D	.9933	.9924	.9904	.9840	.9333	.4557
MLP-Mixer1D	.9938	.9930	.9920	.9854	.9335	.6066
MCNN	.9121	.9357	.9831	.9846	.4362	.2032
DS-CNN	.9029	.9299	.9693	.9471	.4772	.1624
ResNet1D	.8975	.9073	.9450	.9623	.3704	.2188
LeNet1D	.8428	.8505	**.8672**	.9005	.6124	.2437

Table 4: Paired Wilcoxon signed-rank tests, clean-trained vs. augmented, pairs = model $\times$ seed ($n = 24$), Holm-corrected.

metric	clean-trained	+aug	Δ	p_{Holm}
clean	.9949	.9405	-0.0543	3.6×10^{-5}
10 dB	.6988	.9672	$+0.2684$	4.8×10^{-7}
5 dB	.4517	.9695	$+0.5178$	3.6×10^{-5}
0 dB	.2379	.6871	$+0.4492$	4.8×10^{-7}

10 dB but costs 12.7 clean points; $c = 8$ at the same k adds $+49.6$ points for 0.7 clean points; widening to $k = 128$ adds another 2.7 points at 10 dB for 512 more parameters, so the kernel width doubles while the backbone grows by 0.2%. A multi-channel wide stem is therefore the cheapest architectural change we found for a narrow-kernel backbone: 2,608 parameters, or 1.1% of the ResNet1D backbone. The gain is largest where the backbone is most brittle ($+49.6$ for ResNet1D at $c = 8$, $+52.3$ with $k = 128$; $+49.8$ for DS-CNN) and smallest where the backbone already mixes the whole window ($+13.3$ for ViT1D).

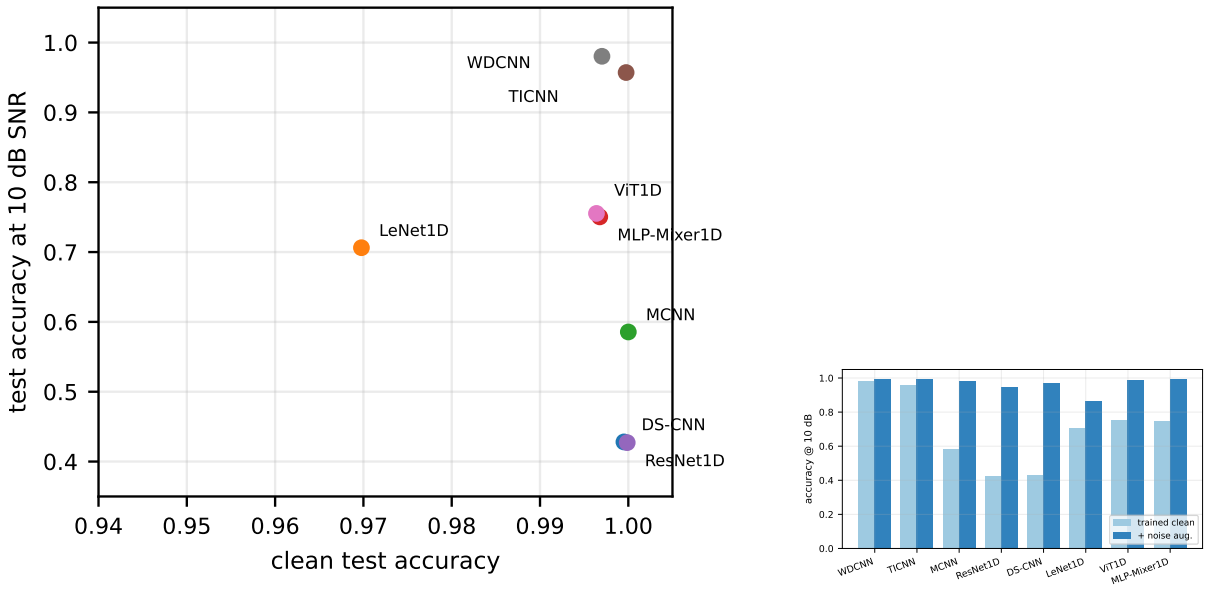


Figure 3: Left: clean accuracy versus 10 dB accuracy for the eight models trained on clean data. The ordering is not preserved: the fitted trend across models is weakly negative. Right: augmentation narrows the 10 dB gap for every model; LeNet1D remains the outlier at 0.87.

Table 5: Wide-kernel front-end ablation (clean training, noise only at test time, one seed, 15 epochs). Δ relative to the same backbone without a front-end. Best value in each accuracy column in bold; the clean maximum is shared and left unbolded.

config	params	clean	10 dB	5 dB	0 dB	Δ10 dB
ResNet1D plain	241,322	1.0000	.4534	.2060	.1000	0
ResNet1D $k64c1$	241,388	.8728	.7959	.6463	.3690	+34.3
ResNet1D $k64c4$	242,258	.9739	.9231	.7907	.4556	+47.0
ResNet1D $k64c8$	243,418	.9929	.9496	.7910	.3955	+49.6
ResNet1D $k32c8$	243,162	.9918	.8799	.5619	.3433	+42.7
ResNet1D $k128c8$	243,930	.9854	.9765	.8799	.4966	+52.3
DS-CNN plain	69,642	.9996	.4224	.2780	.2131	0
DS-CNN $k64c8$	71,290	.9799	.9201	.5896	.2224	+49.8
MCNN plain	184,554	1.0000	.5974	.3097	.1011	0
MCNN $k64c8$	185,418	.9948	**.9884**	**.9332**	**.5343**	+39.1
ViT1D plain	139,850	.9963	.7720	.4858	.2127	0
ViT1D $k64c8$	143,706	.9918	.9049	.7474	.3489	+13.3

4.5 Combining the two mechanisms

The two interventions act on different parts of the pipeline, so they need not be alternatives. Table 6 crosses them: for four representative backbones we train with neither intervention, with augmentation only, with a stem only, and with both (stem $k64c8$, and $k128c8$ for ResNet1D; three seeds, 20 epochs). The augmented and stem rows are re-trained inside this grid, so the augmented rows differ from Table 3 by up to six points (the augmentation draw depends on the seed; Section 6), and the stem rows use 20 epochs here against 15 in Table 5.

Finding 6. The combination is the best configuration at 10 dB for three of the four backbones (ResNet1D 0.989, DS-CNN 0.973, MCNN 0.995) and loses to augmentation alone in the fourth (ViT1D 0.978 vs. 0.991). Two effects make the combination better than either intervention alone on the three narrow-kernel backbones. First, the front-end cancels most of the clean-accuracy cost of augmentation:

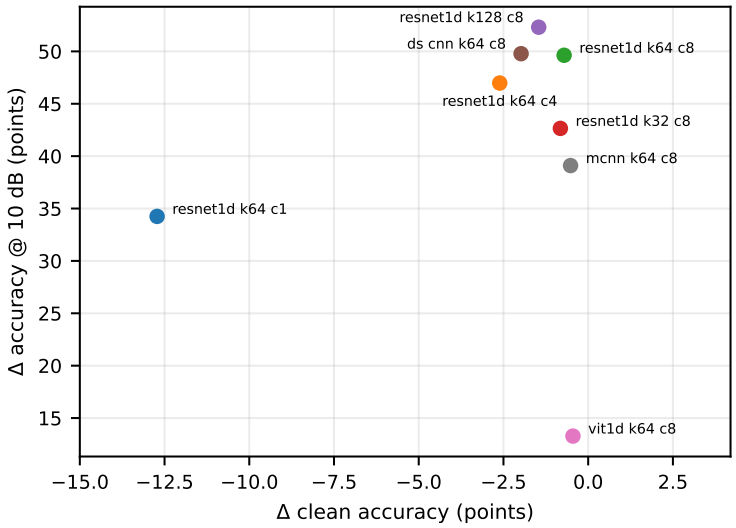


Figure 4: Trade-off of the wide multi-channel front-end: change in 10 dB accuracy against change in clean accuracy, relative to the same backbone without a stem. All configurations lie in the upper-left quadrant, gaining at 10 dB at some cost in clean accuracy: at most 2.6 points for the multi-channel stems, but 12.7 points for the single-channel variant ($k64c1$), which gains 34.3.

augmentation alone costs 8.0–14.6 clean points on these backbones, while stem plus augmentation costs 0.6–4.3, plausibly because the parallel band-pass filters absorb the distribution shift that the noise schedule introduces. Second, the largest gains appear at low SNR: at 0 dB, moving from plain to stem plus augmentation adds 73, 49, 81 and 61 points respectively, and even at -5 dB (outside the augmentation range) the gain remains 10–43 points (not shown in Table 6).

ViT1D behaves differently. Its patch embedding and self-attention already mix information across the window, so the front-end adds only 15 points at 10 dB in this 20-epoch grid (13 in the 15-epoch ablation of Table 5), less than the 24 points gained from augmentation. For this backbone the training recipe matters more than the input layer.

4.6 Open-set behaviour: high closed-set accuracy, poor rejection

We withhold three classes from training and evaluate on a test set that contains all ten, using two holdout configurations: the three 0.021-in faults (classes 3, 6, 9) and the three 0.007-in faults (classes 1, 4, 7). Accuracy on the known classes stays at 0.98–1.00, so closed-set accuracy gives no warning; the failure shows up only in what the model does with the three classes it has never seen.

Table 7 reports rejection quality and Figure 5 plots it against noise accuracy. Averaged over the 48 runs, AUROC is 0.90 (range 0.70–1.00) and TPR at 5% FPR is 0.58 (range 0.01–1.00). Here TPR at 5% FPR is the fraction of known faults still accepted once 5% of the unknown faults are let through; at that threshold the average run also rejects 42% of the known-class samples, and the worst run rejects 99%. The spread across holdout sets, rather than the mean, is the deployment risk: the same architecture can sit near either end depending on which classes are withheld.

Finding 7. Rejection quality depends on the architecture, is unstable across holdout sets, and is *anti*-correlated with noise robustness. WDCNN, the most noise-robust model of the eight (0.98 at 10 dB), has the lowest TPR at 5% FPR (0.12) and the second-lowest AUROC (0.84); LeNet1D has the lowest AUROC (0.785) but a much higher TPR (0.44). ResNet1D and DS-CNN, the two most brittle models in noise, reject best (AUROC 0.97). Across the eight models, Spearman $\rho = -0.64$ ($p = 0.09$, $n = 8$) links 10 dB accuracy and mean AUROC: the association is moderate, not significant, and in the opposite direction to what deployment practice assumes. A plausible explanation is visible in the scores: a model trained to tolerate a wide range of perturbations learns a wide decision region, so an unknown fault lands inside a known class; a brittle model tends to answer with a confidently wrong but outlying prediction. A

Table 6: Four-quadrant comparison of the front-end and noise augmentation (clean training vs. SNR $\sim U[0, 15]$ per batch; three seeds). Δ in points relative to the plain backbone. Best 10 dB and 0 dB value in each model block in bold.

model	condition	clean	10 dB	5 dB	0 dB	Δclean / Δ10 dB
ResNet1D	plain	1.000	.427	.265	.136	—
	+aug	.889	.917	.996	.428	-11.1 / $+49.0$
	stem	.989	.973	.840	.463	-1.1 / $+54.6$
	stem+aug	.986	**.989**	.987	**.865**	-1.4 / $+56.2$
DS-CNN	plain	1.000	.428	.297	.228	—
	+aug	.853	.944	.993	.481	-14.6 / $+51.6$
	stem	.974	.931	.699	.269	-2.6 / $+50.3$
	stem+aug	.957	**.973**	.976	**.716**	-4.3 / $+54.4$
MCNN	plain	1.000	.586	.251	.100	—
	+aug	.920	.980	.982	.419	-8.0 / $+39.5$
	stem	.995	.989	.933	.488	-0.6 / $+40.3$
	stem+aug	.994	**.995**	.992	**.911**	-0.6 / $+41.0$
ViT1D	plain	.996	.755	.468	.271	—
	+aug	.994	**.991**	.983	**.935**	-0.3 / $+23.6$
	stem	.992	.905	.712	.358	-0.5 / $+14.9$
	stem+aug	.976	.978	.976	.885	-2.0 / $+22.3$

Table 7: Open-set rejection: three of ten classes withheld, two holdout configurations $\times$ three seeds. The best mean AUROC, and the best and the worst mean TPR, are bold.

	AUROC		TPR at 5% FPR		mean	
model	3,6,9	1,4,7	3,6,9	1,4,7	AUROC	TPR
DS-CNN	.979	.965	.888	.860	**.972**	**.874**
ResNet1D	.967	.971	.788	.871	.969	.829
MCNN	1.000	.922	.998	.617	.961	.807
TICNN	.993	.822	.975	.047	.908	.511
ViT1D	.958	.841	.776	.352	.900	.564
MLP-Mixer1D	.988	.762	.951	.068	.875	.509
WDCNN	.829	.845	.096	.145	.837	**.121**
LeNet1D	.818	.751	.544	.332	.785	.438

claim that a model is robust in the noise sense should therefore come with an open-set number, because the two properties do not move together. Nor does the augmentation of Section 4.3 fix the other axis: the same per-batch augmentation that adds 27 points at 10 dB lowers mean AUROC from 0.899 to 0.797 with the softmax read-out (Section 4.7; Section 4.8 revisits this).

4.7 Can the robustness recipe also buy rejection?

Section 4.3 recommends per-batch noise augmentation, which adds no parameters and is worth 27 points at 10 dB for 5.4 clean points. If robustness and rejectability were two sides of the same smoothing effect, the same one-line change should improve rejection too. We repeated both holdout configurations with augmented training (32 runs, two seeds, everything else as in Section 4.6).

Table 8 reports the result: rejection does not improve. Mean AUROC falls from 0.899 to 0.797 and mean TPR at 5% FPR from 0.570 to 0.388, and only 4 of the 32 paired (model, holdout, seed) configurations

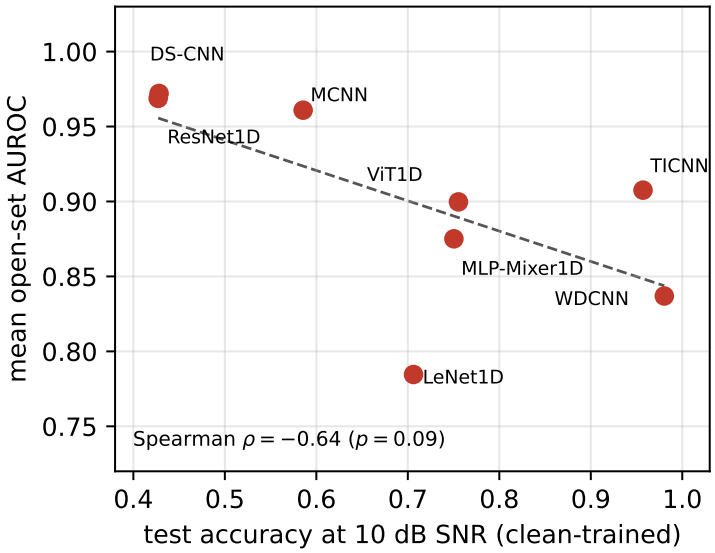


Figure 5: Noise robustness and open-set rejection are not the same property. Each point is one architecture: test accuracy at 10 dB SNR under clean training (horizontal) against mean open-set AUROC over both holdout configurations (vertical). The two are anti-correlated (Spearman $\rho = -0.64$, $p = 0.09$): the model that degrades least under noise (WDCNN) has the lowest TPR at 5% FPR.

Table 8: Open-set rejection under noise-augmented training (32 runs, two seeds). The clean column is the two-seed paired baseline (a subset of the three-seed means in Table 7); Δ is the paired difference; the last row averages the eight models. Best value in each of the four metric columns is bold.

	mean AUROC			mean TPR at 5% FPR		
model	clean	+aug	Δ	clean	+aug	Δ
WDCNN	.858	.822	−0.036	.158	.173	+0.015
TICNN	.904	.833	−0.071	.509	.338	−0.170
MCNN	**.974**	.846	−0.128	**.878**	.541	−0.337
ResNet1D	.965	**.857**	−0.108	.793	**.548**	−0.245
DS-CNN	.968	.644	−0.324	.863	.173	−0.690
LeNet1D	.763	.743	−0.020	.385	.463	+0.078
ViT1D	.887	.829	−0.058	.484	.393	−0.091
MLP-Mixer1D	.873	.798	−0.075	.490	.473	−0.017
mean	.899	.797	−0.103	.570	.388	−0.182

improve in AUROC. A paired Wilcoxon signed-rank test gives $p = 5 \times 10^{-7}$ for AUROC and $p = 6 \times 10^{-3}$ for TPR, i.e. 1×10^{-6} and 6×10^{-3} after Holm correction over the two metrics. Only WDCNN (+0.015) and LeNet1D (+0.078) improve in TPR at 5% FPR, the two whose clean-trained rejection was already weakest; neither improves in AUROC (−0.036, −0.020). The model that loses most is DS-CNN, the best rejector of the three-seed clean-trained set (AUROC 0.972, Table 7): on the paired runs its AUROC falls from 0.968 to 0.644, while its accuracy on the seven known classes at 0 dB is 0.62. With the softmax read-out, then, the augmentation that repairs noise accuracy degrades rejection.

The mechanism is the one described in Section 5.3: augmentation widens the region a class occupies in score space. That is what makes the model tolerant to perturbation, and it is also what lets an unseen fault fall inside a known class. Training and evaluation therefore pull the two objectives in opposite directions, and a practitioner who needs both has to decide at the decision layer rather than in the recipe. Section 4.8 tests whether the decision layer can supply what the recipe removes.

4.8 The rejection gap is mostly in the scoring rule

Section 4.7 measured rejection with the softmax maximum. That is a read-out of a trained network rather than part of it: two networks with the same class geometry can rank unknown faults differently depending on how their confidence is summarised, and smoothing operations such as noise augmentation flatten the softmax distribution in particular. We therefore repeated the whole open-set grid—eight models, two holdout sets, two seeds, clean and augmented, 64 runs—and scored every trained network three ways after training: the softmax maximum (the baseline of Sections 4.6–4.7), the negative free energy $T \log \sum_j e^{z_j/T}$ (we use $T = 1$), and the negative minimum Mahalanobis distance to class-conditional Gaussians fitted on the training logits with a shared covariance. None of the three adds a parameter or a training step; they differ only in how the logit vector is summarised.

Table 9 reports the outcome. Energy does not help: it matches or slightly trails the softmax maximum (mean AUROC 0.889 against 0.899 clean-trained, $p_{\text{Holm}} = 0.06$; 0.785 against 0.796 augmented). The distance-based score does help: mean AUROC rises from 0.899 to 0.953 on clean-trained models (26 of the 32 paired configurations improve, $p_{\text{Holm}} = 1.3 \times 10^{-4}$) and from 0.796 to 0.940 on augmented ones (31 of 32, $p_{\text{Holm}} = 2 \times 10^{-9}$); mean TPR at 5% FPR rises from 0.570 to 0.785 and from 0.387 to 0.746. The augmentation penalty measured in Section 4.7 therefore shrinks by almost an order of magnitude: 0.103 AUROC and 0.182 TPR with the softmax maximum, 0.013 and 0.039 with the Mahalanobis score.

Why does this happen? A softmax maximum measures where a point sits relative to the decision boundary, so any smoothing of the decision surface—exactly what augmentation trains—lowers it. The class-conditional structure in logit space survives the smoothing, and normalising the distance by the spread each class acquires during training keeps unknown faults far from all class centres. Despite its lower average AUROC, the augmented model scored with Mahalanobis (0.940) still beats the clean-trained model scored with the softmax maximum (0.899) on the same networks. The trade-off of Section 4.7 is thus partly a property of the read-out rather than of the features: the information that separates unseen faults is still there after augmentation, and the maximum-probability summary loses it.

One caveat remains: the gain could belong to the coordinate system rather than to the read-out. Logits are a lower-dimensional projection of the features, and a distance in a projected space need not behave like a distance in the space it projects from. We therefore repeated the grid with the same estimator fitted to the input of the last linear layer (penultimate features, 32–256 dimensions across the eight models, obtained by one uniform rule with no architecture-specific hook). Mean AUROC is 0.952 clean-trained and 0.938 augmented: indistinguishable from the logit-space values (paired over 32 configurations: -0.001 and -0.002 AUROC, $p = 0.95$ and $p = 0.64$) and still far above the softmax maximum (0.899 and 0.796), and the augmentation penalty is again 0.014 AUROC instead of 0.103. The score therefore measures distance rather than probability mass, and the coordinates in which it is measured change little. A second caveat is that the score is a free choice in the evaluation, so an open-set number means little without it: the same networks that look damaged by augmentation under one score look almost unaffected under another, just as a robustness claim means little without naming the noise family. A reconstruction-based score remains untested.

Finding 8. Rejection quality depends on the score as much as on the architecture. The same trained networks move from 0.899 (softmax, clean) and 0.796 (softmax, augmented) to 0.953 and 0.940 mean AUROC when the rejection score is a logit-space Mahalanobis distance, with no retraining and no extra parameters, and the cost of noise augmentation on rejection falls from 0.103 to 0.013 AUROC. A fourth read-out, the same distance in penultimate-feature space, reproduces both numbers (0.952 and 0.938), so the gain comes from measuring distance rather than from logit coordinates. A claim that an intervention breaks or preserves rejectability must name the score it was measured with, just as a robustness claim must name the noise family (Finding 9).

4.9 Other noise families

White Gaussian noise is the default test perturbation in this literature, but real bearing signals also carry impulsive (transient), harmonic (electrical interference) and drift (sensor or thermal) disturbances. Table 10 repeats the 10 dB and 0 dB evaluation with three such families, using the four models that span the range

Table 9: Decision-layer scores on the same trained networks (eight models, two holdout sets, two seeds; each cell averages four runs). softmax is the maximum softmax probability used in Sections 4.6–4.7, energy is the negative free energy $T \log \sum_j e^{z_j/T}$, and Mahalanobis is the negative minimum distance to class-conditional Gaussians fitted on the training logits with a shared covariance. All three are computed after training and add no parameters; a fourth read-out, the same distance in penultimate-feature space, is reported in the text. The best score in each row is bold for AUROC and for TPR separately.

		AUROC			TPR at 5% FPR		
model	training	softmax	energy	Mahalanobis	softmax	energy	Mahalanobis
WDCNN	clean	.858	.858	**.938**	.158	.210	**.670**
	aug	.822	.819	**.933**	.173	.224	**.604**
TICNN	clean	.904	.898	**.969**	.508	.503	**.827**
	aug	.833	.808	**.962**	.337	.296	**.774**
MCNN	clean	.974	.961	**.982**	.878	.794	**.928**
	aug	.846	.855	**.972**	.540	.604	**.887**
ResNet1D	clean	.965	.941	**.979**	.793	.650	**.905**
	aug	.857	.831	**.970**	.548	.490	**.839**
DS-CNN	clean	**.968**	.937	.968	**.863**	.694	.858
	aug	.644	.628	**.932**	.172	.154	**.775**
LeNet1D	clean	.763	.794	**.946**	.385	.523	**.746**
	aug	.743	.713	**.952**	.463	.390	**.779**
ViT1D	clean	.887	.882	**.907**	.484	.455	**.641**
	aug	.829	.834	**.882**	.393	.451	**.576**
MLP-Mixer1D	clean	.873	.841	**.932**	.490	.428	**.705**
	aug	.798	.792	**.917**	.473	.353	**.734**
mean	clean	.899	.889	**.953**	.570	.532	**.785**
	aug	.796	.785	**.940**	.387	.370	**.746**

of our results; the families are defined in Section 3.4 and the clean-trained Gaussian column is repeated for reference.

The four families order the models differently. At 10 dB, broadband Gaussian noise is the *hardest* of the four families for the two narrow-kernel CNNs, though not for the other two: under impulsive noise ResNet1D and DS-CNN reach 0.970 and 0.995 at 10 dB, i.e. 54 and 57 points above their own Gaussian accuracy, and at 0 dB DS-CNN keeps 0.929 against 0.228. Sparse transients are statistically closer to the impulsive excitation of a bearing fault than broadband noise is, so suppressing the transients would also suppress the fault signature. *Low-frequency drift*, by contrast, is the one family that separates the models sharply: ViT1D falls 28 points below its Gaussian level (0.474) and DS-CNN reaches only 0.545, while WDCNN loses nothing at 10 dB (0.993) and ResNet1D gains 33 points. This repeats the band-pass reading of Section 4.4: a wide first kernel removes slow wander, while a narrow one maps it onto the fault bands. The ordering induced by the four families is also different in every case: DS-CNN is the best model under impulsive noise and among the worst under drift, WDCNN is first under harmonic and drift interference but is beaten under impulsive noise (second at 10 dB, third at 0 dB), and the Gaussian order (WDCNN $>$ ViT1D $>$ DS-CNN $>$ ResNet1D) survives in none of them at 10 dB.

Finding 9. A single Gaussian probe does not identify a robust model. Which noise family hurts most depends on the architecture—drift for ViT1D, broadband Gaussian for the two narrow-kernel CNNs, and for WDCNN no single family is decisive (0.952–0.993 across the four)—the ranking it induces changes with the family, and a robustness claim that does not name the family cannot be compared across papers. We also note one instability that a single seed would hide: WDCNN under drift at 0 dB averages 0.60 with a standard deviation of 0.33 across two seeds, so its apparent robustness there is not reliable.

Table 10: Architectures under four noise families (clean training, noise at test time only, mean over 2 seeds for the non-Gaussian families and 3 seeds for Gaussian). Accuracy at 10 dB and 0 dB SNR. Best value in each column in bold.

	10 dB SNR				0 dB SNR			
model	Gaussian	harmonic	impulse	drift	Gaussian	harmonic	impulse	drift
WDCNN	**.980**	**.952**	.980	**.993**	**.405**	**.412**	.447	**.604**
ResNet1D	.427	.632	.970	.758	.136	.183	.444	.183
DS-CNN	.428	.611	**.995**	.545	.228	.197	**.929**	.188
ViT1D	.755	.745	.887	.474	.271	.258	.754	.184

Table 11: Parameters and inference latency on the training accelerator (Apple M5, MPS, 1,024-sample windows; mean of 30 (batch 1) / 10 (batch 256) forward passes after warm-up). Latencies were measured while other training jobs shared the device, so absolute numbers are 2–3× pessimistic; relative comparisons are unaffected. Best throughput in bold.

model	params	1×1024 (ms)	256×1024 (ms)	ms/window	windows/s
LeNet1D	3,018	0.50	6.65	0.026	38,472
WDCNN	34,410	1.99	2.23	0.009	**114,605**
DS-CNN	69,642	4.21	17.94	0.070	14,271
TICNN	83,914	1.81	5.25	0.021	48,799
ViT1D	139,850	4.59	42.90	0.168	5,968
MCNN	184,554	4.23	51.42	0.201	4,979
ResNet1D	241,322	5.27	31.03	0.121	8,249
MLP-Mixer1D	747,722	3.52	68.21	0.266	3,753

4.10 Efficiency

Table 11 reports parameter counts and measured latency on the same accelerator used for training. The parameter range spans a factor of 250 (3,018 to 747,722), yet at batch size 1 the wall-clock latency spans only 0.5–5.3 ms, and at batch size 256 the per-sample cost is 0.009–0.27 ms, i.e. 3,750–114,600 windows per second. For a monitoring application that processes one window per second per machine, every architecture in this study is over three orders of magnitude faster than needed.

Finding 10. Parameter count predicts neither latency nor robustness. The two most noise-robust models (WDCNN, TICNN) are also the two fastest at large batch (115k and 49k windows/s), while MCNN, 5× larger than WDCNN and 60× larger than LeNet1D, is slower than both (5k windows/s), and the slowest model of all is the much larger MLP-Mixer1D (3.8k windows/s). Accuracy per parameter, an objective common in lightweight-architecture papers, is therefore the wrong one here: the compute budget is not binding, and what matters is behaviour under noise and on unseen faults.

Table 12 summarises the cost of the interventions instead. Noise augmentation adds no parameters and gains 49 points at 10 dB; the wide stem costs 2,608 parameters (1.1% for ResNet1D) and gains 55; together they gain 56 points at 10 dB and 73 at 0 dB for 1.4 points of clean accuracy. No architecture in our set gains anything comparable for a similar price.

4.11 Second dataset: Paderborn

To check that the conclusions are properties of the evaluation rather than of CWRU, we repeat the noise stress on Paderborn bearing data [21], using the two classes available in the subset we downloaded (healthy, outer race) and the same windowing, split, training and noise pipeline.

We use the 18-bearing KAt (artificial-damage) subset (six healthy bearings K001–K006 and twelve with artificial outer-race damage), 60,000 training and 20,000 test windows, and the same four models and

Table 12: Cost of the interventions studied in this paper, measured in the 2×2 comparison of Section 4.5 on one fixed architecture (ResNet1D, 241,322 parameters, three seeds), except the last row, which applies the same front-end to the other three backbones (four configurations, one seed). Δ columns are differences from the plain model: accuracy in points, parameter count in percent. Largest 10 dB and 0 dB gain in bold.

intervention	Δparams	Δclean	Δ10 dB	Δ0 dB
noise augmentation	0	-11.1	$+49.0$	$+29.3$
wide stem $k128c8$	$+1.1\%$	-1.1	$+54.6$	$+32.7$
stem + augmentation	$+1.1\%$	-1.4	**+56.2**	**+72.9**
front-end, other backbones (4 configs)	$+0.5$ to $+3\%$	-0.5 to -5.0	$+13$ to $+50$	$+1$ to $+43$

Table 13: Paderborn (two classes, 60k/20k windows), mean±std over two seeds. *plain*: trained clean; *aug*: trained with SNR $\mathcal{U}[0, 15]$ per batch. Noise is added at test time only. Best value of each column within each training mode in bold.

model	training	clean	10 dB	5 dB	0 dB
WDCNN	plain	.962±.002	.899±.000	.843±.002	.789±.007
	aug	**.896**±.005	**.941**±.005	.937±.002	**.814**±.017
TICNN	plain	.976±.000	**.908**±.003	**.851**±.012	**.807**±.019
	aug	.826±.083	.924±.049	.944±.018	.800±.040
ResNet1D	plain	**.990**±.000	.885±.024	.769±.005	.558±.013
	aug	.856±.019	.936±.007	**.951**±.008	.811±.001
DS-CNN	plain	.974±.001	.838±.010	.675±.032	.517±.006
	aug	.822±.026	.884±.030	.922±.012	.769±.017

noise pipeline, with and without SNR $\mathcal{U}[0, 15]$ augmentation (Table 13).

Clean accuracy is again compressed, though at a different level: the four plain models span 0.962 to 0.990, i.e. 3 points. Under noise they separate: at 10 dB the plain models fall to 0.838–0.908 (7 points), at 5 dB to 0.675–0.851 (18 points) and at 0 dB to 0.517–0.807 (29 points). The absolute loss is smaller than on CWRU—Paderborn is a harder two-class problem where even the best clean model stays below 99.5%—but the pattern is the same, and on the harder condition models that look equally good clean differ by 29 points.

The rank inversion reappears in a weaker form (plain rows). ResNet1D is the best model clean (0.990) and drops to third at 10 dB (0.885) and to second-worst at 0 dB (0.558, 23 points below WDCNN); DS-CNN is third clean (0.974) and last at both 10 dB (0.838) and 0 dB (0.517). The clean and noisy orderings of the four plain models agree on three of the six pairs and disagree on the other three (Spearman $\rho = 0$). The inversion is therefore directional rather than complete: WDCNN, the worst model clean, is second at 10 dB, but TICNN, second clean, is first at both 10 dB and 0 dB, so we cannot claim a full reversal on this dataset.

Augmentation behaves as it does on CWRU, and it is not free: it costs 6.6 points of clean accuracy for WDCNN and 13.4–15.2 points for the other three, and adds 1.6–5.1 points at 10 dB, 9.3–24.7 at 5 dB and 25.2–25.3 at 0 dB for ResNet1D and DS-CNN. The most informative result is that the four augmented models (eight runs) have the *same* clean and noisy ordering (WDCNN $>$ ResNet1D $>$ TICNN $>$ DS-CNN at both 10 dB and clean), i.e. the inversion disappears once the perturbation is part of the training distribution. We note that TICNN with augmentation has a large seed spread (clean 0.826 ± 0.083) and loses 0.8 points at 0 dB, so the ordering of the other three models is more stable.

Finding 11. The collapse under noise and the failure of clean accuracy to predict it reproduce on a second rig (two datasets, two label spaces, two sampling rates): the clean-best model falls to third at 10 dB and to second-worst at 0 dB, and the clean-worst model is not the noisy-worst. The inversion is weaker than on CWRU—TICNN leads in both conditions—but it disappears under augmentation in the same way, so the phenomenon is a property of clean training rather than of one dataset.

5 Discussion

5.1 Why the benchmark cannot see the difference

CWRU clean test accuracy has been at the ceiling for years, and every model in our set sits within 3 points of every other. A metric that is bounded above by 100% and already lies at 97–100% has almost no dynamic range left: the differences that remain are small in absolute terms, the ordering within the top group changes from seed to seed, and changing one protocol (leaky $\rightarrow$ cross-load) moves accuracy 2–5 times further than the entire spread of the eight models. The conclusion is not that the models are equally good; it is that *clean CWRU accuracy has stopped carrying information*. A number that every method has reached cannot be evidence of progress.

5.2 Two different robustness mechanisms

Our ablation separates two mechanisms that the literature often mixes together. The first is a training-distribution effect: exposing the model to noisy inputs makes it less sensitive to them, and this alone is worth $+26.8$ points at 10 dB averaged over eight architectures. The second is an architectural effect: a wide multi-channel front-end maps the raw window through a bank of learnable band-pass filters before the backbone sees it, which is worth up to $+52$ points for a narrow-kernel backbone and only 13 points for a backbone that already mixes the whole window (Table 5). The two are complementary but not equivalent: on the same backbone the stem costs 1.1 points of clean accuracy where augmentation costs 11.1, the stem adds 1.1% of the parameters, and augmentation is free in parameters but still helps the models the stem cannot fix.

That a wide first kernel helps under noise is not new (WDCNN made it standard in 2017), but our results show that the effective part of the design is the number of parallel channels more than the kernel width. A single 64-sample kernel adds only 66 parameters and gains $+34$ points at 10 dB but destroys clean accuracy, because one filter has to serve every fault frequency. Eight parallel filters at the same width cover several resonance bands at once, which is what the clean-accuracy recovery in Table 5 measures. Widening the kernel further adds little: $k = 128$ adds another 2.7 points for 512 more parameters, 0.2% of the backbone.

There is a physical reading of this. At 12 kHz, 64 samples span 5.3 ms; the ball-pass frequencies of the 6205-2RS drive-end bearing at 1,797 r.p.m. are 107 Hz (BPFO, 9.3 ms) and 162 Hz (BPFI, 6.2 ms) [1]. A single wide kernel is comparable to one period of the fault signature, so it can act as a matched filter for one resonance band only; a bank of them covers several. Noise, which is broadband, is suppressed by the band-pass character while the periodic fault signature passes.

5.3 Robustness and rejectability are different axes

Our open-set audit adds a further misalignment to the rank inversion of Section 4.2. A model can be robust to additive noise—because its front end averages over many samples, or because it was trained to ignore perturbation—and still be unable to tell a fault it has never seen from one it has seen. The two properties are moderately anti-correlated in our data ($\rho = -0.64$, $p = 0.09$, $n = 8$): WDCNN, the model with the widest effective receptive field and the most noise-tolerant decision function, is also the one that maps unknown faults most strongly into the interior of a known class; TICNN collapses on one holdout (TPR 0.05) but not the other (0.98), so the effect is at least partly holdout-specific.

There is no contradiction here. A noise-tolerant classifier is by construction a smoothing operator: it is trained to give the same answer for many perturbations of the input, and an unknown fault class is just one more perturbation that falls inside the smoothed region. Rejection quality instead depends on how *peaked* the score function is around the training classes, which is what narrow, brittle models have. The practical consequence is that the two requirements have to be evaluated, and optimised, separately: calling a noise-robust model deployment-ready without an open-set number mixes up two different axes of behaviour.

How much of the second axis is the score is a separate question, and Section 4.8 answers it: with a Mahalanobis distance score the same networks reach 0.953 (clean) and 0.940 (augmented) mean AUROC (0.952 and 0.938 when the distance is measured in penultimate-feature space), and the cost of augmentation falls from 0.103 to 0.013 AUROC in logit space (0.014 in penultimate-feature space). The trade-off of Section 4.7 is therefore mostly a property of the softmax read-out: smoothing provides noise tolerance but flattens the maximum-probability summary, while the class-conditional geometry that separates unseen faults survives it. The practical rule is the same as for robustness claims: report the open-set number together with the score that produced it.

5.4 What a practitioner should do

Given a fixed budget and a narrow-kernel backbone, our results suggest a concrete order: (i) add per-batch noise augmentation, which is free in parameters and worth 27 points on average; (ii) add a wide multi-channel stem, which is worth up to 52 points in the controlled ablation (55 in the combined grid) for about 1% more parameters; (iii) only then consider architectural novelty. Both are small changes to the training loop and the first layer; Appendix A gives the reproduction commands.

5.5 Implications for evaluation practice

A reviewer cannot detect a 55-point robustness gap from a table of clean accuracies, and an author may have little incentive to report a stress test that might show a smaller number. The fix is structural rather than technical: noise stress and open-set evaluation should be mandatory in the reporting checklist, as some vision venues now do. We give a seven-item list in Appendix A.

5.6 Relation to prior critique

Recent position papers argue that vibration benchmarks are contaminated by overlapping-window leakage and by splits that are not comparable [6, 7]. Our audit agrees on the protocol problem but finds the effect small: at most 0.63 points at full data and at most 0.92 in a 20-per-class regime. The more serious problem is saturation: when every method scores 99%, the protocol can no longer rank methods because the metric has no resolution, although it still shifts absolute accuracy by 5–13 points. Our contribution is to show what a resolution-restoring test looks like and what it reveals.

6 Limitations

The scope of the study is limited in several ways.

Datasets. Both datasets come from laboratory test rigs with a handful of bearings, constant speed and controlled loads. The Paderborn subset we use contains artificial damage on the outer race only, so it supports a two-class healthy/outer task rather than a full three-class diagnosis; it tests the evaluation method, not the label space. Real machines have varying speed, time-varying load, multiple sensors and long-term drift. Noise is additive and white in our main matrix; we check three other families, but we do not simulate sensor faults, missing samples or non-stationarity.

Models. Our set of eight architectures is a sample, not a census. We leave out transformer hybrids, graph networks and domain-adaptation methods, and we deliberately do not tune per-architecture hyperparameters, so we cannot claim that any architecture is at its best. Because the harness is identical, the differences we measure are not explained by unequal tuning effort; per-architecture tuning could still change them.

Noise protocol. SNR is defined per window after z-score normalisation, with a fixed noise realisation per evaluation; we do not average over noise realisations, which would narrow the error bars. The augmentation schedule (uniform 0–15 dB per batch) is one of many, and we did not search it.

Open set. We use two held-out-class configurations (three of ten), and the rejection score in the main audit is the softmax maximum. We compare three alternatives on the same networks (Section 4.8): the negative free energy behaves like the softmax maximum, while a Mahalanobis distance improves mean AUROC from 0.899 to 0.953 clean-trained and from 0.796 to 0.940 augmented in logit space, and to 0.952 and 0.938 in penultimate-feature space, leaving an augmentation cost of 0.013–0.014 AUROC; a reconstruction-based score remains untested. The large spread across holdout configurations (TICNN reaches 0.05 and 0.98 TPR at 5% FPR under the two configurations, and across models the range is 0.01–1.00) also means that a single open-set number is a sample, not a property.

Randomness in augmentation. The noise schedule draws a fresh SNR per mini-batch. In the released code the draw is derived from the seed, so a run is exactly reproducible; the augmented results reported here were produced before that change, so re-running an augmented configuration reproduces it to within 1–8 points at 10 and 5 dB (the random draws differ). All augmented numbers in this paper are means over two or three seeds, and seed-to-seed variation is part of what those means average over. The claims we draw from them are differences of 20–50 points.

Statistical scope. Two to three seeds, one data ordering, one hardware platform. Paired tests control for model and seed but not for the choice of split, which we leave to future work as a variance component.

7 Conclusion

The standard benchmark has saturated, and the usual robustness probe is not comparable across papers. We re-evaluated eight lightweight 1D architectures under one leakage-safe harness and found that clean accuracy is uninformative while noise accuracy separates the same models by 55 points. Much of the separation is a property of the training recipe, which a one-line per-batch noise augmentation repairs; a multi-channel wide front-end recovers up to 52 points for about 1% more parameters, with the number of channels the dominant factor, although kernel width also matters. Open-set rejection varies from 0.12 to 0.87 TPR at 5% FPR across architectures even when closed-set accuracy is near 100%, and the most noise-robust model has the lowest TPR at 5% FPR—an axis that the augmentation repairing noise accuracy pushes the other way (AUROC 0.90 to 0.80) unless the rejection score is chosen with it, in which case almost all of that loss is recovered (0.80 to 0.94 with a Mahalanobis distance score, in logit or penultimate-feature space). In practice, clean CWRU accuracy should not be reported as evidence of improvement, a noise matrix and an open-set number should accompany every claim of robustness, and a change to the training recipe should be tried before a change of architecture.

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Conflict of interest

The author declares no competing interests.

Data and code availability

All splits, per-seed result files, figures and the exact commands that regenerate every table and figure in this paper are available at `https://github.com/jinchi2007/clean-acc-not-diagnosis`. CWRU and Paderborn data are public and are not redistributed.

A Reproduction commands and reporting checklist

Reproduction. From the released repository, the following commands regenerate the paper (seeds and epochs are fixed in the scripts unless a flag sets them):

1. `python -m code.data.cwru` — build the four split protocols and verify zero sample overlap.
2. `python -m experiments.grid --grid e2_noise_clean --seeds 0,1,2 --epochs 20`
3. `python -m experiments.grid --grid e4_aug --seeds 0,1,2 --epochs 20`
4. `python -m experiments.grid --grid e3_openset --seeds 0,1,2 --epochs 20`
5. `python -m code.models.stem_ablation --part 0 --nparts 2`, then `--part 1 --nparts 2`, then `--merge` — the wide-stem ablation.
6. `python -m experiments.e5_combo --part 0 --nparts 2`, then `--part 1 --nparts 2`, then `--merge` — the four-quadrant comparison.
7. `python -m experiments.e6_noise_types` — harmonic, impulsive and drift noise.
8. `python -m experiments.e7_efficiency` — parameters and latency.
9. `python -m experiments.grid --grid e0_sat --seeds 0,1 --epochs 20` — the four split protocols used in Table 1.
10. `python experiments/e0_fewshot.py --seeds 0,1` — the few-shot leakage audit.
11. `python -m experiments.e8_paderborn` — the second dataset (Paderborn).
12. `python -m experiments.e9_openset_aug --seeds 0,1 --epochs 20` — rejection under augmented training (Section 4.7).
13. `python -m experiments.e10_scores --seeds 0,1 --epochs 20` — the three rejection scores (Section 4.8).
14. `python -m experiments.e11_penultimate --seeds 0,1 --epochs 20` — the same grid scored in penultimate-feature space.
15. `python -m experiments.stats && python -m experiments.figures` — tables and figures.
16. `python experiments/check_tables.py` — re-verifies every table in this manuscript cell by cell against `results/*.json`.

A single CWRU run takes a few seconds to about two minutes (2–8 min on Paderborn) on a laptop-class accelerator (Apple M5, MPS backend); the full matrix is about 430 runs, i.e. a few hours.

Reporting checklist. We suggest that every vibration-diagnosis paper report:

1. the split unit (window or recording) and the number of shared sampling points between train and test (ideally zero, verified);
2. clean accuracy *and* accuracy at two or more SNRs, with the noise model and the SNR definition;
3. the number of seeds (three where feasible), with mean and standard deviation, and paired statistics for any claimed improvement;
4. parameter count and inference latency at the intended batch size;

5. open-set behaviour when the deployment scenario can encounter unseen faults (AUROC and TPR at a fixed FPR);

6. whether the training set was noise-augmented, and the augmentation schedule;

7. code and split files, or an explicit statement that they are unavailable.

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